Question Paper Preview

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Industry 4.0

Section Id :	640653124755
Number of Questions :	22
Section Marks :	45
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531864780 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536040503. ✔ YES

6406536040504. ✖ NO

Question Id : 6406531864781 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (2 to 3)

Question Label : Comprehension

GreenLeaf Chemicals Pvt. Ltd. manufactures liquid fertilizers. One of the main raw materials required is *Chemical X*. The production department has reported the following annual usage details:

• Annual requirement of Chemical X: 12,000 liters
• Cost of placing a purchase order: ₹150 per order
• Annual holding (carrying) cost per liter: ₹2
Due to limited warehouse capacity, the management wants to optimize inventory levels and reduces unnecessary ordering costs. The purchase manager has suggested calculating the Economic Order Quantity (EOQ) to determine the best ordering policy.

As the newly appointed Inventory Analyst, you are asked to calculate the given subquestions:

Sub questions

Question Number : 2 Question Id : 6406531864782 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the Economic Order Quantity (EOQ) in liters/order

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1300 to 1350

Question Number : 3 Question Id : 6406531864783 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the number of purchase orders to be placed per year?
[Assume 1 year=52 weeks]

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

9

Question Number : 4 Question Id : 6406531864784 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Colony Shop is stocking heavy coats for next winter. Colony pays \$50 for coat and sells it for \$110. At the end of the winter season, Colony Shop offers the coats at \$55 each. The demand for coats during the winter season is more than 20 but less than or equal to 30, all with equal probabilities. Determine the optimal order quantity that will maximize revenue for Colony Shop. Consider demand to be continuous.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

29 to 33

Question Id : 6406531864785 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (5 to 6)

Question Label : Comprehension

Suppose you need to accept reservation for a flight. There are total 100 seats in the aircraft. You have two classes of passengers - economy and business. One economy seat gets you 2,500 Rs of revenue while a business seat gets 7,500 Rs. From your experience, you know the demand for business travelers follows a uniform distribution in [10,40]. Then answer the given subquestions:

Sub questions

Question Number : 5 Question Id : 6406531864786 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the Booking limit?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

70

Question Number : 6 Question Id : 6406531864787 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the protection level?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 7 Question Id : 6406531864788 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Match the following columns:

Industry / Application (Col A)	Best-Suited Layout Type (Col B)
A1. Automobile Assembly Line	B1. Process Layout
A2. Hospital (Diagnostic Department)	B2. Fixed-Position Layout
A3. Shipbuilding Yard	B3. Cellular Layout
A4. Manufacturing Small Batch Custom Products	B4. Product Layout

Options :
6406536040510. ✖ A1–B1, A2–B2, A3–B3, A4–B4
6406536040511. ✖ A1–B3, A2–B1, A3–B4, A4–B2
6406536040512. ✔ A1–B4, A2–B1, A3–B2, A4–B3
6406536040513. ✖ A1–B4, A2–B4, A3–B1, A4–B2

Question Number : 8 Question Id : 6406531864789 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Select the correct form of internet from column B corresponding to the example in Column A.

Column A	Column B
A1. Smart metering	B1. Consumer
A2. Banking system	B2. Commercial
A3. Entertainment integrated devices	B3. Industrial
	B4. Private

Options :
6406536040514. ✖ A1: B3, A2: B2, A3:B4
6406536040515. ✔ A1: B3, A2: B2, A3:B1
6406536040516. ✖ A1: B2, A2: B3, A3:B1

Question Number : 9 Question Id : 6406531864790 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

A factory works for two 8-hour shifts each day and produces a total of 24,000 electric bulbs using a set of workstations. There are 8 activities required to manufacture the bulb, such that the sum of all activity times is equal to 12 seconds. How many workstations are required to main the specified level of production (assuming that activities can be combined into those workstations in any order) (enter your answer in numeric value)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 10 Question Id : 6406531864791 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Calculate the Euclidean distance between two machines located at the coordinates A(5,10) and B(2,6) in a shop floor.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.8 to 5.3

Question Number : 11 Question Id : 6406531864792 Question Type : MSQ

Correct Marks : 1.5 Max. Selectable Options : 0

Question Label : Multiple Select Question

There are four demand points at A(1,2), B(2,3), C(3,5), D(4,1). The respective demand at these points are 7, 1, 3, 5. Select the optimal location based on cross-median approach.

Options :

6406536040519. ✔ (2,2)

6406536040520. ✔ (3,2)

6406536040521. ✔ (2.5,2)

6406536040522. ✖ (2,3)

Question Number : 12 Question Id : 6406531864793 Question Type : MSQ
Correct Marks : 1 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following example(s)s can be referred as seasonal pattern?
Options :
6406536040523. ✖ Economic recission
6406536040524. ✔ Increases Ceiling Fan sales during winters
6406536040525. ✔ Increased tourist footfall towards the end of the year

Question Number : 13 Question Id : 6406531864794 Question Type : MCQ
Correct Marks : 1.5
Question Label : Multiple Choice Question
Which of the following functions, Logistics Management does not cover?
Options :
6406536040526. ✖ Supply Management
6406536040527. ✖ Materials Management
6406536040528. ✔ Customer Management
6406536040529. ✖ Distribution Management

Question Number : 14 Question Id : 6406531864795 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are considered in the Vehicle Routing Problem but not considered in TSP?
Options :
6406536040530. ✖ Subtour elimination
6406536040531. ✔ Any capacity restrictions
6406536040532. ✔ A specific starting point
6406536040533. ✖ Distance travelled restriction

Question Number : 15 Question Id : 6406531864796 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
What is the name of the forecasting method represented by the following set of equations? The notations represent the usual meaning

$$F_{t+1} = a_t + b_t$$

Where
$$a_t = \alpha D_t + (1-\alpha)(a_{t-1} - b_{t-1});$$
$$b_t = \beta(a_t - a_{t-1}) + (1-\beta)b_{t-1}$$

Options :
6406536040534. ✖ Exponential smoothing
6406536040535. ✖ Winter's model for seasonality
6406536040536. ✔ Holte's method for trend
6406536040537. ✖ None of these

Question Number : 16 Question Id : 6406531864797 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following supply chain costs is/are affected by distribution network structure?
Options :
6406536040538. ✔ Inventory
6406536040539. ✔ Transportation
6406536040540. ✔ Facilities and handling
6406536040541. ✔ Information
6406536040542. ✖ Price

Question Number : 17 Question Id : 6406531864798 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Which of the following is not an example of digitization?
Options :
6406536040543. ✖ Scanning a document to get the PDF document
6406536040544. ✖ Converting a physical 2D sheet to CAD drawing
6406536040545. ✖ Converting physical invoice to e-copy
6406536040546. ✔ Creating an automated report from the existing cost data

Question Id : 6406531864799 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (18 to 20)

Question Label : Comprehension

A contractor estimates that the size of the workforce needed over the next 5 weeks is 5, 7, 8, 4 and 6 workers, respectively. Excess labour kept on the force will cost $300 per worker per week, and new hiring in any week will incur a fixed cost of $400 plus $200 per worker per week. The objective is to minimize the total cost of labour needed for the project. If Dynamic Programming is applied to this problem, answer the given subquestions:

Sub questions

Question Number : 18 Question Id : 6406531864800 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How is a stage represented by?

Options :

6406536040547. ✔ Week

6406536040548. ✖ Number of labours available in a week

6406536040549. ✖ Number of new hires in a week

6406536040550. ✖ Total cost of labour in a week.

Question Number : 19 Question Id : 6406531864801 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What are the alternatives in a stage?

Options :

6406536040551. ✖ Week

6406536040552. ✖ Number of new hires in a week

6406536040553. ✔ Number of labours in a week

6406536040554. ✖ Number of labours available at the end of previous week

6406536040555. ✖ Total cost of labour in a week.

Question Number : 20 Question Id : 6406531864802 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the optimal solution (objective function value) in Stage 5 when the state variable takes value 4?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

8

Question Id : 6406531864803 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (21 to 22)

Question Label : Comprehension

The owner of a newsstand wants to determine the number of newspapers of *USA Now* to be stocked at the start of each day. The owner pays 30 cents for a copy and sells it for 75 cents. The sale of the newspaper typically occurs between 7:00 and 8:00 am. Newspaper left at the end of the day are recycled for an income of 5 cents a copy. How many copies should the owner stock every morning, assuming that the demand for the day can be described as:

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 21 Question Id : 6406531864804 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

How many copies should the owner stock each morning, assuming that the daily demand follows a normal distribution with a mean of 300 copies and a standard deviation of 20 copies?
[Standard_Norm_inv(critical fractile)=0.643]

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

305 to 315

Question Number : 22 Question Id : 6406531864805 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

How many copies should the owner stock each morning, assuming that the daily demand follows a discrete probability distribution, f(D), defined as follows?

D	200	220	300	320	340
f(D)	0.1	0.2	0.4	0.2	0.1

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

300

Question Number : 23 Question Id : 6406531864806 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the Industry 4.0 version of transportation.

Options :

6406536040559. ✖ IC engine

6406536040560. ✖ Electric car

6406536040561. ✔ Hyperloop

Question Number : 24 Question Id : 6406531864807 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Geet Mobile manufactures mobile phones of various types. Their product portfolio comprises of 15 types of headsets. These headsets comprise of limited types of components. Only a few of the components needs customization for different headsets. Which facility layout will best suit for Geet Mobile’s production?

Options :

6406536040562. ✖ Product layout

6406536040563. ✔ Group layout

6406536040564. ✖ Fixed position layout

6406536040565. ✖ All of these

Question Number : 25 Question Id : 6406531864808 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statements from below:

Options :

6406536040566. ✔ Optimal solution in a facility location problem using Euclidean method rarely matches the center of gravity

6406536040567. ✖ Optimal solution in a facility location problem using Euclidean method will result in a region

6406536040568. ✔ Optimal solution in a facility location problem using Euclidean method will result in a point

6406536040569. ✖ Optimal solution in a facility location problem using Euclidean method matches the centre of gravity in not of the cases

Question Number : 26 Question Id : 6406531864809 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

When is outsourcing not useful?

Options :

6406536040570. ✔ When the (production) volumes are large

6406536040571. ✔ Uncertainty in supply

6406536040572. ✔ If the service requires specific resources rather than service

6406536040573. ✔ Quality is impacted

Question Number : 27 Question Id : 6406531864810 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Pranjal is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk-powder packets in the neighbouring shelves. He picked a few milk-powder packets, however these are not part of his list. He was searching for Good-day biscuits. He could identify only one packet in the shelf remaining. However, there were lot of Sun-fist biscuits lying in the next shelf. So he picked ten Sunfist packets. Now map the scenarios with the items below:

Product	Scenario
P1)Tea:Coffee	S1)Complementary
P2) Tea: milk-powder packet	S2)Substitution
P3) Good-day biscuits: Sun-fist bicuit	S3)Inventory dependant demand
	S4)Impulse buying

Options :

6406536040574. ✖ P1-S1, P2-S3, P3-S4

6406536040575. ✔ P1-S2, P2-S1, P3-S3

6406536040576. ✖ P1-S2, P2-S1, P3-S4